Dots and Boxes

Verbose Rules and Design Notes for Rectangular and Custom-Shaped Mobile Web Versions

This guide explains the standard rules, scoring, turn handling, edge cases, and recommended behavior for a customizable phone-friendly web app version of Dots and Boxes. It covers both ordinary rectangular grids and custom shapes made from selected boxes.

1. Game summary

Dots and Boxes is a turn-based territory game for two or more players. The board begins as a set of dots connected by empty possible line segments. On each turn, a player draws one available line between two adjacent dots. If that line completes one or more boxes, the player claims those boxes, scores points, and takes another turn. The game ends when no lines remain. The player with the most claimed boxes wins.

2. Components and vocabulary

Term	Meaning
Dot	A point on the board. Legal lines connect adjacent dots horizontally or vertically.
Line / edge	A segment between two adjacent dots. A line is either undrawn or drawn. Once drawn, it cannot be erased or reused.
Box / cell	A square area bounded by four possible edges. A box exists only if it is part of the playable shape.
Completed box	A playable box whose top, right, bottom, and left edges have all been drawn.
Owner	The player who completed and claimed a box.
Move	One action of drawing one legal undrawn edge.
Extra turn	The additional turn earned when a player completes at least one box.
Chain	A sequence of boxes where taking one box may force or enable a run of additional box captures.
Playable shape	The set of boxes included in the board. This can be a rectangle or a custom selection of boxes.

3. Standard board setup

A standard rectangular Dots and Boxes board is usually described by the number of boxes, not by the number of dots. For example, a 4 by 3 board has 4 boxes across and 3 boxes high. That board needs 5 columns of dots and 4 rows of dots.

Example: a 4 by 3 box grid. Letters show claimed boxes.

●	—	●	—	●	—	●	—	●
|	A	|		|		|		|
●	—	●	—	●	—	●	—	●
|		|		|	B	|		|
●	—	●	—	●	—	●	—	●
|		|		|		|		|
●	—	●	—	●	—	●	—	●

4. Rectangular custom sizes

For a rectangular board, let width be the number of boxes across and height be the number of boxes down. The dot grid is always one larger in each direction: dot columns = width + 1 and dot rows = height + 1.

Quantity	Formula
Boxes	width x height
Dots	(width + 1) x (height + 1)
Horizontal possible edges	width x (height + 1)
Vertical possible edges	(width + 1) x height
Total possible moves	width x (height + 1) + (width + 1) x height

Recommended app limits: use small default boards on phones, such as 3 x 3, 4 x 4, 5 x 5, or 6 x 6. Larger boards work, but they need zooming, scrolling, or larger touch targets. A good minimum touch target for selectable edges is about 40 CSS pixels, even if the visible line is much thinner.

5. Custom-shaped boards

A custom-shaped board can be created by letting the user draw, tap, or select boxes on an invisible rectangular backing grid. The playable board is then the set of selected boxes. A box is playable if it exists in that selected set. Empty spaces are holes or outside areas and should not be scored.

The easiest representation is a two-dimensional boolean matrix: true means the box exists, false means it does not. For example, a 5 x 4 backing grid might contain an L-shape, a cross, a donut with a hole, or any arbitrary island of boxes.

6. Lines on custom shapes

For custom shapes, an edge is playable if it borders at least one playable box. This rule lets players draw boundary edges around the outside of the shape and edges between neighboring playable boxes. Edges that touch no playable boxes should not be shown or selectable.

An edge can border one or two playable boxes. Boundary edges border one playable box. Interior edges border two playable boxes. Drawing one edge can complete one box, or in rare cases two boxes at once if it is the shared final side between two nearly completed boxes.

7. Legal moves

- A legal move draws exactly one currently undrawn playable edge.
- The edge must connect two adjacent dots horizontally or vertically.
- The same edge cannot be drawn twice.
- A player may draw a boundary edge, an interior edge, or any other playable edge, even if it gives the next player an advantage.
- Diagonal lines are not legal in the standard game.
- A line that is not adjacent to any playable box is not part of the board and should not be selectable.

8. Turn sequence

1. The current player selects one legal undrawn edge.
2. The app marks that edge as drawn using the current player color, a neutral color, or a completed-board style.
3. The app checks every playable box adjacent to that edge.
4. Any adjacent box whose four edges are now drawn becomes completed.
5. If one or more boxes were completed, the current player claims all of them, scores that many points, and immediately takes another turn.
6. If no box was completed, the turn passes to the next player.

9. Capturing boxes

A player captures a box only when their move causes the box to become fully surrounded. It does not matter who drew the earlier three sides. The player who draws the fourth side owns the box.

If a single edge completes two boxes at the same time, the player claims both boxes and scores two points. The player still receives only one continuing turn, but that turn is earned because at least one box was completed.

10. Extra-turn rule

The extra-turn rule is the main reason Dots and Boxes is more strategic than it first appears. Whenever a player completes at least one box, they do not give up the turn. They continue playing until they make a move that completes no boxes, or until the game ends.

This means a player may capture a long chain of boxes in a single turn. The app should keep the active player unchanged after every scoring move. Do not alternate turns after a completed box.

11. Scoring and winning

- Each captured box is worth 1 point unless you intentionally create a variant with different scoring.
- The score equals the number of boxes owned by each player.
- The game ends when every playable edge has been drawn. Equivalently, every playable box should be completed, unless the shape includes no boxes.
- The player with the highest score wins.
- If two or more players tie for the highest score, the game is a draw among those players.

12. Two-player and multiplayer rules

The traditional game is for two players, but the rules work for three or more players. Use a fixed turn order, such as Player 1, Player 2, Player 3, then back to Player 1. A player who scores keeps taking turns, so the next player in order only acts after a non-scoring move.

For online or pass-and-play modes, clearly show whose turn it is. On small screens, use persistent score chips at the top or bottom of the screen and a large turn indicator.

13. Custom shape validity recommendations

- Allow any non-empty set of playable boxes if you want creative shapes.
- For a cleaner competitive mode, require the selected boxes to be connected through shared edges.
- Decide whether diagonal-only contact counts as connected. For standard Dots and Boxes, edge-connected shapes are easier to understand than diagonal-connected shapes.
- Holes are allowed if the app treats their surrounding edges as playable boundary edges.
- Disconnected islands are playable, but they can feel odd because players may jump between islands on any turn. If allowed, explain that any legal edge anywhere on the board may be selected.

14. Implementation model for rectangular and custom boards

A reliable implementation separates boxes from edges. Store boxes as cells and edges as unique objects. Each box references four edge IDs: top, right, bottom, and left. Each edge references the one or two boxes it borders.

Object	Suggested fields
Box	id, row, col, exists, owner, topEdge, rightEdge, bottomEdge, leftEdge
Edge	id, orientation, row, col, drawn, drawnBy, adjacentBoxIds
Player	id, name, color, score
Game	players, currentPlayerIndex, boxes, edges, moveHistory, status

For rectangular boards, generate all boxes and all edges using width and height. For custom shapes, first generate only the selected boxes. Then create an edge whenever one side of a selected box needs it. If a neighboring selected box shares that side, both boxes should reference the same edge object.

15. Move validation logic

When the user taps or drags near an edge, convert that input into an edge ID. Before accepting the move, verify that the game is active, the edge exists, the edge is playable, and the edge has not already been drawn.

After drawing the edge, check only the boxes adjacent to that edge. This is faster and less error-prone than scanning the whole board after every move. A box is complete when all four referenced edges are drawn.

16. Recommended pseudo-code

```
function playEdge(edgeId):
  edge = edges[edgeId]
  if game.status !== "active": return invalid
  if !edge or edge.drawn: return invalid

  edge.drawn = true
  edge.drawnBy = currentPlayer.id

  completed = []
  for boxId in edge.adjacentBoxIds:
    box = boxes[boxId]
    if box.exists and box.owner == null and allFourEdgesDrawn(box):
      box.owner = currentPlayer.id
      completed.push(boxId)

  currentPlayer.score += completed.length
  moveHistory.push({ edgeId, playerId: currentPlayer.id, completed })
```

```
if completed.length == 0:
  advanceToNextPlayer()
else:
  keepSamePlayer()

if allPlayableEdgesDrawn():
  game.status = "complete"
  calculateWinner()
```

17. Drawing a custom shape in the app

A practical shape editor can use a normal rectangular backing grid. The user chooses the backing grid size, then taps boxes to toggle them on or off. The playable game board is generated from the selected boxes.

- Step 1: choose a backing grid size, such as 6 boxes wide by 5 boxes high.
- Step 2: display faint selectable cells rather than edges.
- Step 3: let the user tap, drag, paint, erase, mirror, rotate, or clear cells.
- Step 4: validate the selected shape according to your chosen rules.
- Step 5: generate the edge graph from the selected cells.
- Step 6: start the game using only generated playable edges.

18. Phone-specific interaction guidance

- Make the tap target around each edge larger than the visible line.
- Highlight the edge under the finger before confirming the move.
- Consider requiring release on the same highlighted edge to avoid accidental moves.
- Use haptics or a brief animation when a box is captured.
- Show claimed boxes with initials, icons, or colors. Do not rely on color alone; include text or pattern indicators for accessibility.
- Support portrait mode first. Landscape can be a bonus for large boards.
- For large or custom boards, use pinch zoom or fit-to-screen scaling, but keep touch targets usable.

19. Rule variants you may optionally add

Variant	Description
Misere Dots and Boxes	The player with the fewest boxes wins. This changes strategy dramatically.
Timed turns	Each player has a clock or per-move timer.
No extra turns	Players alternate every move even after completing boxes. This is simpler but less traditional.
Bonus boxes	Some boxes are worth 2 or more points.
Blocked edges	Some edges cannot be drawn, creating maze-like

	boards.
Power moves	Rare abilities such as drawing two edges, undoing one edge, or claiming a neutral bonus. These should be optional because they change the classic strategy.

20. Common mistakes to avoid

- Describing a board by dots when the user expects boxes. A 5 by 5 dot grid creates only 4 by 4 boxes.
- Ending the turn after a completed box. In the standard game, scoring gives another turn.
- Creating duplicate edge objects for the same shared side. Adjacent boxes must share the same interior edge.
- Letting players tap edges that touch no playable box in a custom shape.
- Failing to award two boxes when a shared edge completes both at once.
- Making edge tap targets too small for phones.
- Using only color to identify box ownership. Add initials, numbers, or symbols too.

21. Suggested default settings

Setting	Recommendation
Default mode	Two players, pass-and-play
Default rectangular size	4 x 4 boxes
Beginner sizes	2 x 2, 3 x 3, 4 x 4
Competitive sizes	5 x 5 and 6 x 6
Custom shape editor	Tap boxes on a backing grid to include or remove them
Custom shape validation	Require at least one playable box; optionally require edge-connected shape
Scoring	1 point per claimed box
Turn rule	Scoring player continues

22. Plain-English rule text for your app

You may use the following text directly in your game help screen:

> *Players take turns drawing one line between two neighboring dots. If your line completes a box, you claim that box, score 1 point, and take another turn. If your line does not complete a box, the turn passes to the next player. A line can be drawn only once. The game ends when all playable lines have been drawn. The player with the most claimed boxes wins. On custom boards, only boxes included in the shape count, and only lines touching at least one included box are playable.*

23. Final checklist for your build

- Can the user choose width and height for rectangular boards?
- Can the user create a custom shape by selecting boxes?
- Do custom shapes generate correct boundary and shared edges?
- Does the scoring player keep the turn?
- Can one move claim two boxes?
- Does the game detect completion only after all playable edges are drawn?
- Are tap targets large enough on a phone?
- Does the UI clearly show current player, score, captured boxes, and legal moves?
- Does the game handle ties?
- Does the help screen explain the difference between dots, boxes, and custom shapes?

Prepared as a rules and implementation guide for a customizable mobile web app version of Dots and Boxes.